

TU/CSE

1. What are the limitations of ASCII? 3
2. What is the importance of Unicode? Name two representation schemes for Unicode values. 3 + 2
3. What is the significance of Nyquist-Shannon Sampling Theorem? What are the apparent approximations in digital representation of speech? 4 + 3

4. If a system has to record audio of 1 minute where the highest frequency is known to be 5 KHz, what would be the size of the digital representation? What parameter value is to be assumed in this computation? 3 + 2

5. Give two examples each of inflection and derivation in English morphology. 2

6. If we see the sequence “I want to make a phone, we can guess that the next word would be “call. What insight does this provide about the capability to estimate the validity or probability of a word sequence in a language? 3